

Dreamhome



A Performance Innovative Task

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INTRODUCTION

DreamHome is a web-based Property Management System built to help real estate agencies manage their day-to-day operations efficiently. It centralizes the management of branches, properties, owners, clients, staff, lease agreements, property viewings, and inspections into a single, role-protected application.

The system replaces manual or spreadsheet-based record-keeping with a structured relational database backed by a modern Laravel web application. It is designed to support multiple office branches, each with its own staff and property listings, while providing managers and administrators a consolidated view of all organizational data.

DreamHome addresses common pain points in property management such as tracking property availability, scheduling viewings, creating and monitoring lease contracts, and recording routine property inspections — all within a secure, authenticated environment.

SYSTEM OBJECTIVES

General Objective

To develop a secure, full-stack web-based property management system that enables real estate agencies to manage properties, staff, clients, owners, leases, viewings, and inspections through a role-based, multi-branch platform built with Laravel and MySQL.

Specific Objectives

- Implement a role-based access control system with four roles: Admin, Manager, Supervisor, and Staff — each with appropriate permissions to resources.
- Provide full CRUD (Create, Read, Update, Delete) operations for all major entities: Branches, Properties, Owners, Clients, Staff, Leases, Viewings, and Inspections.
- Allow managers and administrators to create and manage lease agreements, linking clients to specific properties with defined terms and payment details.
- Enable scheduling and tracking of property viewings where staff can accompany clients to visit available properties.
- Support routine property inspection recording with date, assigned staff, and comments.
- Build a real-time dashboard that summarizes key metrics including total properties, available units, active leases, registered clients, and recent activity logs.
- Enforce data integrity through database-level foreign key constraints and application-level validation rules.
- Provide a clean, responsive user interface using Tailwind CSS, accessible via desktop and mobile browsers.

SCOPE AND LIMITATIONS

Scope: The DreamHome system covers the following functional areas:

- Branch management — maintaining office locations, contact information, and organizational structure.
- Property listings — recording and filtering rental properties by type, rent, availability, owner, and branch.
- Owner management — maintaining records of property owners and their contact details.
- Client registration — recording prospective tenant information including preferred property type and maximum rent.
- Staff management — maintaining employee records, positions, salaries, and supervisor relationships.
- Lease management — creating and tracking rental agreements including duration, monthly rent, deposit, payment method, and linked parties.
- Viewing management — scheduling and logging property viewings between clients and staff.
- Inspection management — recording property inspections carried out by staff with observations.
- Authentication and authorization — user login/logout and role-based middleware protecting all routes.
- Dashboard reporting — an overview page displaying aggregate counts and recent activity.

Limitations

- The system does not include an online tenant portal; clients cannot log in or interact with the system directly.
- Payment processing is not implemented; lease payment tracking is limited to recording payment method and deposit status.
- Document generation (PDF leases, inspection reports) is not included in this version.
- Email notifications and automated reminders are not implemented.
- Image/photo uploads for properties are not supported in this version.
- The system does not include a public-facing property listing page for prospective tenants.

Target Users

DreamHome is designed for use by staff within a multi-branch real estate agency.

The system defines four distinct user roles:

Admin — Full access to all modules. Intended for system-wide oversight; can manage all staff, leases, properties, clients, and branches across all offices.

Manager — Access to all modules. Intended for branch managers who oversee daily operations, approve leases, and manage staff records.

Supervisor — Access to properties, owners, clients, viewings, and inspections. Intended for field supervisors who coordinate property viewings and inspections but do not manage leases or staff.

Staff — Limited CRUD on branches. Intended for general office staff with the most restricted access, used for day-to-day branch-level tasks.

All roles require authentication. Unauthenticated users are redirected to the login page for every protected route.

Tools and Technologies Used

Front-End

- HTML
- CSS
- Bootstrap

Back-End

- Laravel (PHP Framework)
- PHP

Database

- PostgreSQL
- SQL

Development Tools

- Visual Studio Code (VS Code)
- Laragon
- pgAdmin 4
- Composer

Version Control & Collaboration

- GitHub

Database Design & Modeling

- dbdiagram.io
- ERD (Entity Relationship Diagram)

Framework Architecture

- MVC (Model-View-Controller) Architecture

System Features

Authentication Module

The system uses Laravel's built-in authentication with session-based login. Users authenticate with their email and password. The LoginController validates credentials via `Auth::attempt()` and regenerates the session on success. Role-based middleware (`role:admin,manager, etc.`) protects all major route groups. Unauthenticated access to any protected route redirects to `/login`.

Dashboard

The Dashboard module provides an at-a-glance summary of the entire system. It displays eight key metrics: total properties, available properties, total owners, total branches, total staff, total clients, total leases, and total inspections. It also lists the five most recently added properties and the five most recent inspections with their assigned staff.

Branch Management

Branches represent the physical office locations of the agency. Each branch has an address (street, area, city, postcode), telephone number, and fax number. All authenticated users including staff can view and manage branches. Branch records serve as the anchor for properties, clients, and staff assignments.

Property Management

Properties are the core entity of the system. Each property is identified by a unique property number and contains address details, type (e.g. house, flat), number of rooms, monthly rent, and availability status. Properties are linked to an owner and a branch. The property listing page supports filtering by branch, type, maximum rent, and availability status. Full CRUD operations are available to admin, manager, and supervisor roles.

Owner Management

Owners are individuals or entities that own properties listed with the agency. Owner records include first name, last name, address, and telephone number. Each property can be linked to one owner, and owners can hold multiple properties. CRUD access is restricted to admin, manager, and supervisor roles.

Client Management

Clients are prospective tenants registered with the agency. Client records store personal details, preferred property type, maximum acceptable rent, and any comments. Clients are linked to the branch where they registered and the staff member who handled their registration. Clients can be associated with viewings and leases.

Staff Management

Staff records contain full personal and employment details including position, salary, date joined, NIN (National Insurance Number), sex, date of birth, and a self-referencing supervisor field (a staff member can supervise other staff). Staff are assigned to branches. CRUD operations for staff are restricted to admin and manager roles.

Lease Management

The Lease module manages formal rental agreements between a client and a property, handled by a staff member. Each lease record stores start/end dates, duration in months (3–12), monthly rent, deposit amount, deposit paid status, and payment method. Leases are filtered by client or property number on the listing page. This module is accessible only to admin and manager roles.

Viewing Management

Viewings log visits by clients to a specific property, optionally accompanied by a staff member. The composite primary key is (client_no, property_no, view_date), preventing duplicate viewing records for the same combination. Viewings include an optional comments field.

Inspection Management

Inspections record routine checks of properties carried out by staff. Each inspection has an auto-incremented ID, a linked property, an assigned staff member, an inspection date, and comments. Inspections are accessible to all roles with

supervisor-level access or higher. Recent inspections are highlighted on the dashboard.

Database Design

branches

- branch_no — CHAR(4), PRIMARY KEY
- street — VARCHAR(60), NOT NULL
- area — VARCHAR(40), NULLABLE
- city — VARCHAR(30), NOT NULL
- postcode — VARCHAR(10), NOT NULL
- tel_no — VARCHAR(20), NULLABLE
- fax_no — VARCHAR(20), NULLABLE

owners

- owner_no — CHAR(5), PRIMARY KEY
- f_name — VARCHAR(30), NOT NULL
- l_name — VARCHAR(30), NOT NULL
- street — VARCHAR(60), NULLABLE
- city — VARCHAR(30), NULLABLE
- postcode — VARCHAR(10), NULLABLE
- tel_no — VARCHAR(20), NULLABLE

properties

- property_no — CHAR(5), PRIMARY KEY
- street — VARCHAR(60), NOT NULL
- city — VARCHAR(30), NOT NULL
- postcode — VARCHAR(10), NULLABLE
- type — VARCHAR(20), NULLABLE
- rooms — SMALLINT, NULLABLE
- rent — DECIMAL(8,2), NULLABLE
- is_available — BOOLEAN, DEFAULT true
- owner_no — CHAR(5), FK → owners
- branch_no — CHAR(4), FK → branches

clients

- client_no — CHAR(5), PRIMARY KEY
- f_name — VARCHAR(30), NOT NULL
- l_name — VARCHAR(40), NOT NULL
- street, area, city, postcode — VARCHAR, NULLABLE
- tel_no — VARCHAR(20), NULLABLE
- pref_type — VARCHAR(20), NULLABLE
- max_rent — DECIMAL(8,2), NULLABLE
- comments — TEXT, NULLABLE
- registered_by — CHAR(5), NULLABLE (staff reference)
- branch_no — CHAR(4), FK → branches

staff

- staff_no — CHAR(5), PRIMARY KEY
- f_name, l_name — VARCHAR(30), NOT NULL
- sex — CHAR(1), NULLABLE
- dob — DATE, NULLABLE
- nin — VARCHAR(12), NULLABLE
- position — VARCHAR(20), NOT NULL
- salary — DECIMAL(9,2), NULLABLE
- date_joined — DATE, NULLABLE
- branch_no — CHAR(4), FK → branches
- supervisor_no — CHAR(5), FK → staff (self-referencing)

leases

- lease_no — CHAR(5), PRIMARY KEY
- monthly_rent — DECIMAL(8,2), NULLABLE
- payment_method — VARCHAR(40), NULLABLE
- deposit — DECIMAL(8,2), NULLABLE
- deposit_paid — BOOLEAN, DEFAULT false
- date_start, date_end — DATE, NULLABLE
- duration_month — SMALLINT, NULLABLE (3–12 months)
- client_no — CHAR(5), FK → clients
- property_no — CHAR(5), FK → properties
- staff_no — CHAR(5), FK → staff

viewings

- client_no — CHAR(5), PK + FK → clients
- property_no — CHAR(5), PK + FK → properties
- view_date — DATE, PRIMARY KEY (composite with client_no and property_no)
- staff_no — CHAR(5), FK → staff
- comments — TEXT, NULLABLE

inspections

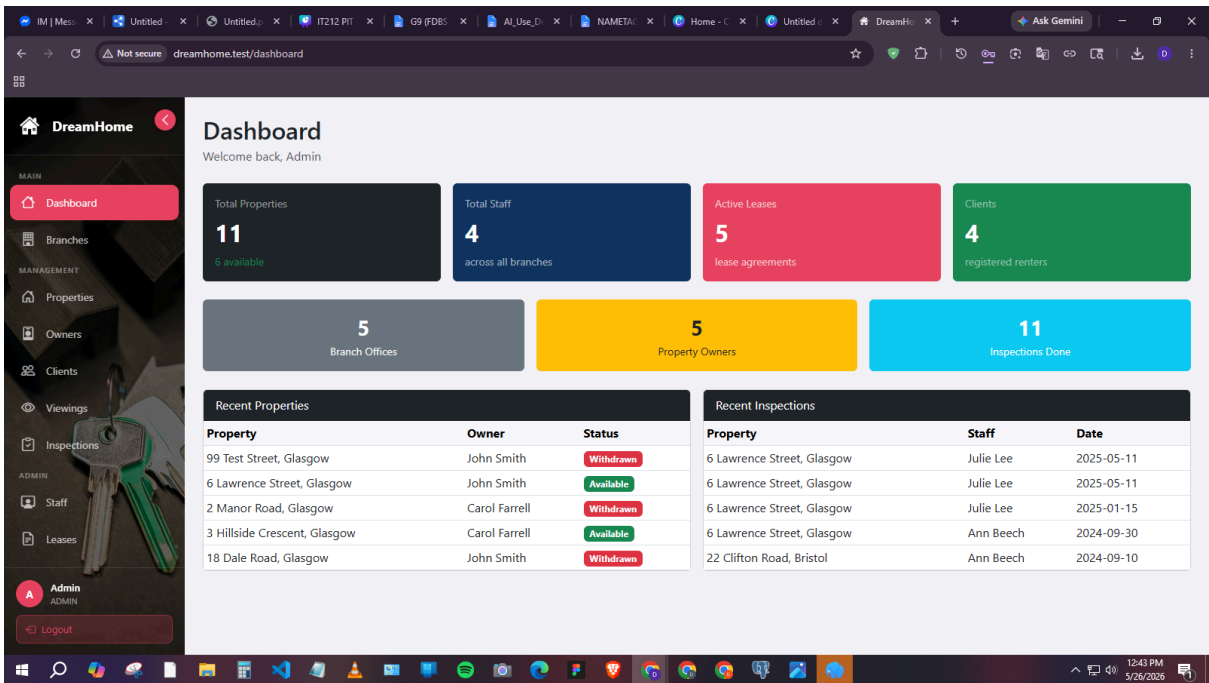
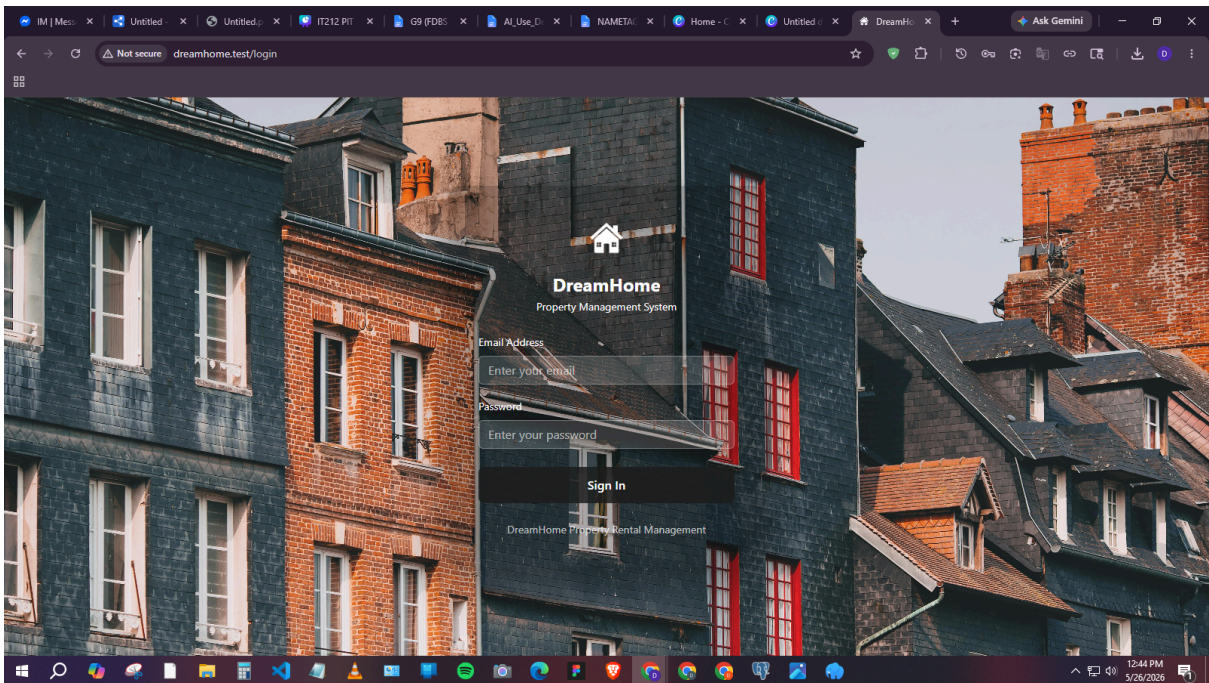
- inspection_id — INT, AUTO INCREMENT, PRIMARY KEY
- property_no — CHAR(5), FK → properties
- staff_no — CHAR(5), FK → staff
- inspection_date — DATE, NOT NULL
- comments — TEXT, NULLABLE

Entity Relationships

- A Branch has many Properties, Staff, and Clients.
- An Owner has many Properties.
- A Property belongs to one Owner and one Branch; it can have many Viewings, Leases, and Inspections.
- A Client belongs to one Branch; a Client can have many Viewings and one active Lease.
- A Staff member belongs to one Branch; a Staff member can have a Supervisor (self-referencing FK). Staff are linked to Viewings, Leases, and Inspections.
- A Lease links one Client, one Property, and one Staff member.
- A Viewing links one Client, one Property, and optionally one Staff member (composite PK).
- An Inspection links one Property and one Staff member.

System Screenshots

The following describes the major pages of the DreamHome application. Screenshots should be captured from a running instance of the application.



DreamHome

MAIN

Dashboard

Branches

MANAGEMENT

Properties

Owners

Clients

Viewings

Inspections

ADMIN

Staff

Leases

Admin

ADMIN

Logout

Branches

Add Branch

Branch No	Street	Area	City	Postcode	Tel No	Fax No	Actions
B001	163 Main Street	Patrick	Glasgow	G11 9QX	0141-339-2178	0141-339-2179	Edit Delete
B002	22 Deer Road	Yoker	Glasgow	G13 1GX	0141-339-3000	0141-339-3001	Edit Delete
B003	472 Scotland Street	Partick	Glasgow	G5 8NW	0141-339-4000	0141-339-4001	Edit Delete
B004	16 Argyll Street	London	London	WC2N 5DU	0171-884-5000	0171-884-5001	Edit Delete
B005	73 Queen Street	Bristol	Bristol	BS1 4LR	0117-929-1000	0117-929-1001	Edit Delete

DreamHome

MAIN

Dashboard

Branches

Properties

MANAGEMENT

Owners

Clients

Viewings

Inspections

ADMIN

Staff

Leases

Admin

ADMIN

Logout

Properties

Add Property

-- All Branches --

-- All Types --

Max Rent

-- All Status --

Filter

Reset

Property No	Street	City	Type	Rooms	Rent	Owner	Branch	Status	Actions
PB001	22 Clifton Road	Bristol	House	4	£700.00	Susan Lee	Bristol	Available	Edit Delete
PB002	5 Redland Grove	Bristol	Flat	2	£480.00	Tina Murphy	Bristol	Withdrawn	Edit Delete
PG001	6 Lawrence Street	Glasgow	Flat	3	£350.00	John Smith	Glasgow	Available	Edit Delete
PG002	2 Manor Road	Glasgow	Flat	3	£375.00	Carol Farrell	Glasgow	Withdrawn	Edit Delete
PG003	18 Dale Road	Glasgow	House	5	£600.00	John Smith	Glasgow	Withdrawn	Edit Delete
PG004	5 Novar Drive	Glasgow	Flat	4	£450.00	Tina Murphy	Glasgow	Available	Edit Delete
PG005	3 Hillside Crescent	Glasgow	Studio	1	£250.00	Carol Farrell	Glasgow	Available	Edit Delete
PG099	99 Test Street	Glasgow	Flat	2	£350.00	John Smith	Glasgow	Withdrawn	Edit Delete
PL001	47 Oakfield Avenue	London	Flat	2	£800.00	James Watson	London	Available	Edit Delete
PL002	12 Kensington Road	London	House	6	£1,500.00	James Watson	London	Available	Edit Delete
PL003	9 Baker Street	London	Flat	3	£950.00	Susan Lee	London	Withdrawn	Edit Delete

DreamHome

MAIN

Dashboard

Branches

MANAGEMENT

Properties

Owners

Clients

Viewings

Inspections

ADMIN

Staff

Leases

Admin

ADMIN

Logout

Viewings

Schedule Viewing

-- All Properties --

-- All Clients --

Filter

Reset

Client	Property	View Date	Staff	Comments	Actions
Laura Cruz	12 Kensington Road, London	2024-04-22	John White	Too far from workplace.	EditDelete
Laura Cruz	22 Clifton Road, Bristol	2024-04-20	Ann Beech	Suitable for family. Interested in leasing.	EditDelete
James Lim	5 Novar Drive, Glasgow	2024-04-13	Julie Lee	Preferred this one over PG002.	EditDelete
James Lim	2 Manor Road, Glasgow	2024-04-11	Ann Beech	Needs more time to decide.	EditDelete
Sarah Tan	12 Kensington Road, London	2024-04-05	John White	Loves the area but rent is too high.	EditDelete
Sarah Tan	47 Oakfield Avenue, London	2024-04-02	John White	Very interested, asked about lease terms.	EditDelete
Mike Ritchie	18 Dale Road, Glasgow	2024-03-26	Julie Lee	Too big for their needs.	EditDelete
Mike Ritchie	6 Lawrence Street, Glasgow	2024-03-24	Ann Beech	Client liked the location, wants to see again.	EditDelete

DreamHome

MAIN

Dashboard

Branches

MANAGEMENT

Properties

Owners

Clients

Viewings

Inspections

ADMIN

Staff

Leases

Admin

ADMIN

Logout

Inspections

Record Inspection

-- All Properties --

-- All Staff --

Filter

Reset

ID	Property	Inspected By	Date	Comments	Actions
12	6 Lawrence Street, Glasgow	Julie Lee	2025-05-11	Annual inspection completed.	EditDelete
11	6 Lawrence Street, Glasgow	Julie Lee	2025-05-11	Annual inspection completed.	EditDelete
9	6 Lawrence Street, Glasgow	Julie Lee	2025-01-15	Ceiling repair completed. Property back in good condition.	EditDelete
2	6 Lawrence Street, Glasgow	Ann Beech	2024-09-30	Cracked ceiling in living room. Requires urgent repair.	EditDelete
8	22 Clifton Road, Bristol	Ann Beech	2024-09-10	Damp patch found in bathroom ceiling. Needs inspection by plumber.	EditDelete
7	12 Kensington Road, London	John White	2024-08-05	Boiler making noise. Maintenance team notified.	EditDelete
6	47 Oakfield Avenue, London	John White	2024-07-20	Property well maintained. No issues.	EditDelete
4	18 Dale Road, Glasgow	Ann Beech	2024-07-01	Crockery needs to be replaced.	EditDelete
5	5 Novar Drive, Glasgow	Carol Dean	2024-06-15	Garden overgrown. Tenant reminded of maintenance duties.	EditDelete
3	2 Manor Road, Glasgow	Julie Lee	2024-05-10	Minor scuff marks on walls. Otherwise clean.	EditDelete
1	6 Lawrence Street, Glasgow	Julie Lee	2024-04-12	No problems found. Property in good condition.	EditDelete

Staff

-- All Branches -- -- All Positions -- Filter Reset

Staff No	Name	Position	Branch	Salary	Supervisor	Date Joined	Actions
SG37	Ann Beech	Manager	Glasgow	£32,000.00	None	2005-09-01	View Edit Delete
SL21	John White	Manager	Glasgow	£35,000.00	None	2000-06-01	View Edit Delete
SL41	Julie Lee	Supervisor	Glasgow	£24,000.00	None	2010-03-15	View Edit Delete
SL52	Carol Dean	Secretary	Glasgow	£18,000.00	Julie Lee	2018-07-01	View Edit Delete

Add Staff

Leases

-- All Clients -- -- All Properties -- Filter Reset

Lease No	Client	Property	Monthly Rent	Start Date	End Date	Duration	Deposit Paid	Actions
LS099	Mike Ritchie	18 Dale Road, Glasgow	£400.00	2025-06-01	2026-06-01	12 months	Yes	View Edit Delete
LS005	Mike Ritchie	2 Manor Road, Glasgow	£375.00	2024-06-15	2024-09-15	3 months	No	View Edit Delete
LS004	Laura Cruz	22 Clifton Road, Bristol	£700.00	2024-06-01	2024-12-01	6 months	Yes	View Edit Delete
LS003	James Lim	5 Novar Drive, Glasgow	£450.00	2024-05-15	2025-05-15	12 months	No	View Edit Delete
LS002	Sarah Tan	47 Oakfield Avenue, London	£800.00	2024-05-01	2024-08-01	3 months	Yes	View Edit Delete

Create Lease

Login Page (/login) — Email and password login form. Displays validation errors on invalid credentials. Redirects authenticated users to the dashboard.

Dashboard (/dashboard) — Summary cards showing counts of properties, available units, owners, branches, staff, clients, leases, and inspections. Includes two recent-activity tables: latest properties added and most recent inspections.

Properties List (/properties) — Filterable table of all properties with columns for property number, address, type, rooms, rent, availability, owner, and branch. Filter bar allows filtering by branch, type, max rent, and status.

Property Create/Edit — Form for adding or updating a property record. Fields include property number, address, type, rooms, rent, availability toggle, owner selection, and branch selection.

Clients List (/clients) — Tabular list of registered clients showing name, contact info, preferred property type, max rent, and branch.

Staff List (/staff) — Table of staff members with name, position, branch, salary, and date joined. Accessible to admin and manager roles only.

Leases List (/leases) — Table of all lease agreements with client name, property address, rent, start/end dates, and staff. Filterable by client or property.

Lease Detail (/leases/{id}) — Full detail view of a single lease showing all fields including deposit status, payment method, and linked parties.

Viewings List (/viewings) — Table of scheduled and completed viewings showing client, property, date, staff, and comments.

Inspections List (/inspections) — Table of property inspections with property address, staff name, date, and comments.

Source Code Explanation

Routing (`routes/web.php`)

All application routes are defined in `routes/web.php` using Laravel resource routes.

Middleware is applied to protect routes based on authentication and user roles:

- **auth** — requires users to log in before accessing the system.
- **role:admin,manager** — restricts staff and lease management features.
- **role:admin,manager,supervisor** — allows access to properties, owners, clients, viewings, and inspections.

Laravel's `Route::resource()` method is used to automatically generate CRUD routes for each module.

Controllers (app/Http/Controllers/)

The system uses multiple controllers to manage different modules:

LoginController handles user authentication and logout.

DashboardController retrieves dashboard statistics using raw SQL queries through Laravel's `DB::select()`.

PropertyController and **LeaseController** implement filtering, searching, and CRUD operations using raw SQL queries such as `DB::select()`, `DB::insert()`, `DB::update()`, and `DB::delete()`.

Other controllers such as **BranchController**, **OwnerController**, **ClientController**, **StaffController**, **ViewingController**, and **InspectionController** follow the same CRUD structure.

Models (app/Models/)

The application uses Eloquent models for defining relationships and managing database entities:

- **Property**
- **Branch**
- **Owner**
- **User**

Although models are used for relationships and validation, most database operations are implemented using raw SQL queries for better query control and complex joins.

Views (resources/views/)

Views are organized into folders based on modules such as properties/, owners/, leases/, clients/, and others. Each module includes:

- **index.blade.php** — displays records and filters
- **create.blade.php** — form for adding records
- **edit.blade.php** — form for updating records
- **show.blade.php** — detailed record display

The layouts/ folder contains the main application layout shared across all pages.

Middleware

A custom role middleware checks the authenticated user's role and restricts access to unauthorized users. If access is denied, the system returns a `403 Forbidden` response.

Testing Summary

Feature Tested	Test Method	Test Data Used	Observed Result
User Login	Manual browser test	Valid credentials (admin@example.com)	Successful redirect to dashboard
Invalid Login	Manual browser test	Wrong password	Error message displayed; no session created
Role Restriction — Staff	Manual browser test	Staff-role user accessing /leases	403 Forbidden response returned
Property CRUD	Manual browser test	New property PG001, 3 rooms, rent 8000	Property created, listed, edited, and deleted successfully
Property Filtering	Manual browser test	Filter by branch B001, type=house	Filtered results returned correctly
Lease Creation	Manual browser test	Client CL001, Property PG001, 6 months	Lease saved; linked client and property reflected
Viewing Duplicate Prevention	Manual browser test	Same client/property/date combination	Duplicate insert rejected by composite PK constraint
Dashboard Counts	Manual browser test	Known seed data set	Counts match actual records in each table
Inspection Listing	Manual browser test	3 inspections in database	All 3 inspections visible in list with staff name
Session Logout	Manual browser test	Click logout	Session invalidated; redirect to login page

Conclusion and Recommendations

Conclusion

DreamHome successfully delivers a functional, role-secured property management system built on the Laravel framework. The system covers all major workflows of a typical real estate agency: managing branch offices, listing and filtering rental properties, registering clients and owners, scheduling viewings, creating lease agreements, and recording property inspections.

The application demonstrates the practical use of Laravel features including route middleware, resource controllers, Blade templating, database migrations, and raw query building. The database schema enforces referential integrity through foreign key constraints, and the application layer adds input validation on all write operations.

Role-based access control is cleanly implemented, restricting sensitive operations (lease management, staff records) to authorized personnel while allowing lower-privilege users access to day-to-day operational views.

Recommendations for Future Development

- Migrate raw `DB::select()` queries to full Eloquent ORM usage for cleaner, more maintainable code and easier unit testing.
- Implement a tenant portal where registered clients can log in, view available properties, request viewings, and check their lease status.
- Add PDF generation for lease agreements and inspection reports using a library such as DomPDF or Snappy.
- Integrate email notifications for upcoming lease expirations, inspection schedules, and new viewing requests.
- Add property photo upload functionality using Laravel Storage and cloud storage (e.g. Amazon S3).

- Implement a public-facing property search page for prospective tenants to browse available listings without authentication.
- Expand the testing suite with feature tests using Laravel's built-in testing tools (PHPUnit / Pest) to cover all CRUD operations and access control rules.
- Add audit logging to track who created, updated, or deleted records and when.

References

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- PHP 8.3 Documentation — <https://www.php.net/docs.php>
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- Composer Documentation — <https://getcomposer.org/doc/>
- GitHub Repository (Source Code) — <https://github.com/dnnvrr2/dreamhome>